

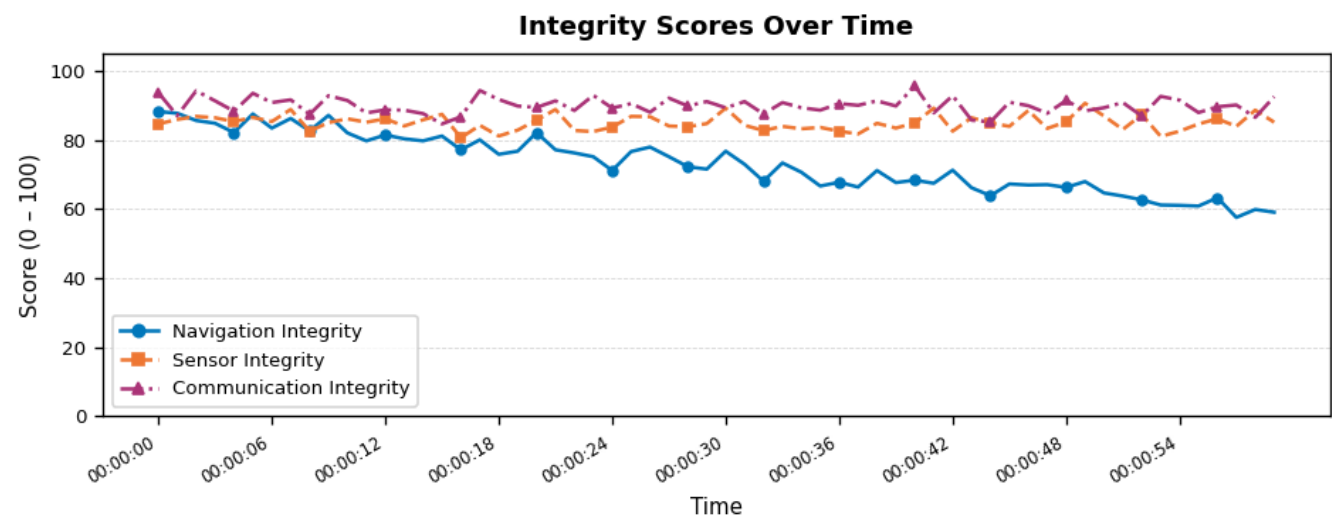
Mission Integrity System (MIS) — V1

Scenario: Gradual Gps Degradation
Generated: 2026-04-20 18:47:07 UTC
Demonstration report generated from simulated data

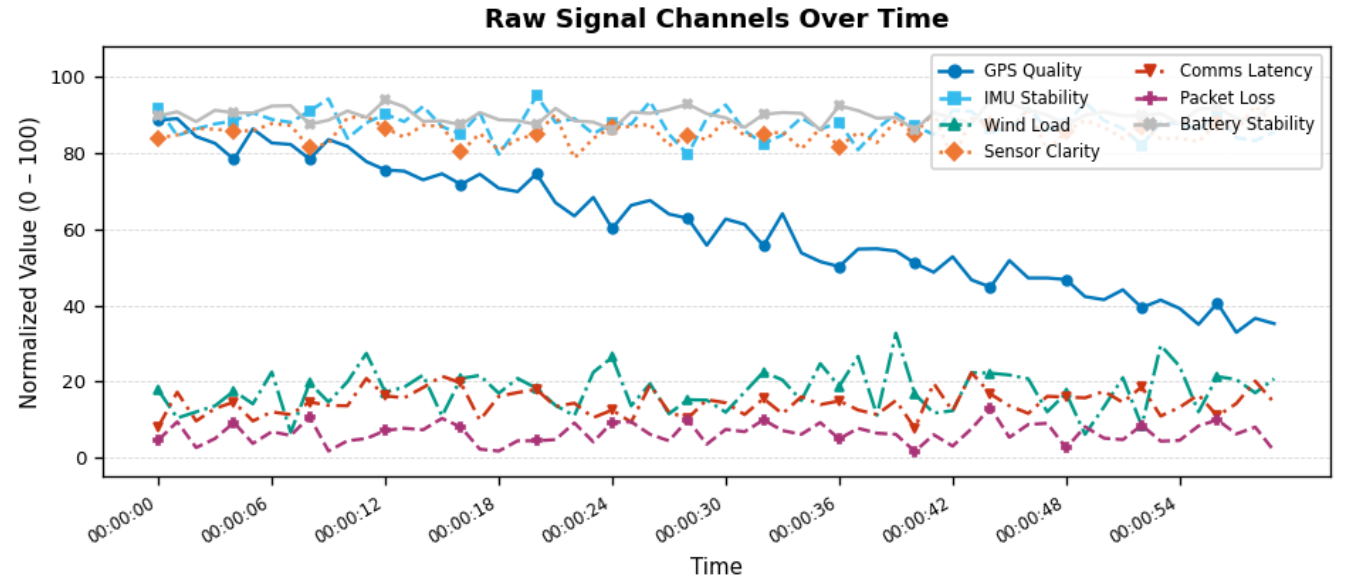
Scenario Summary

Field	Value
Final State	Degraded
Navigation Integrity	59.1 / 100
Sensor Integrity	85.2 / 100
Communication Integrity	92.6 / 100
Causal Explanation	Degraded: Navigation integrity in degraded band (40–59).

Integrity Scores Over Time



Raw Signal Channels Over Time



Event Log

Time	Severity	Type	Message
00:00:00	INFO	State Change	Mission state initialized: Stable.
00:00:24	WARNING	State Change	State transition: Stable → Drift Emerging. Drift Emerging: Navigation integrity in drift-emerging band (60–75).
00:00:24	WARNING	Threshold Breach	Navigation integrity crossed below threshold 75: 75.2 → 71.2.
00:00:25	INFO	State Change	State transition: Drift Emerging → Stable. Stable: all integrity domains within acceptable bounds.
00:00:28	WARNING	State Change	State transition: Stable → Drift Emerging. Drift Emerging: Navigation integrity in drift-emerging band (60–75).
00:00:28	WARNING	Threshold Breach	Navigation integrity crossed below threshold 75: 75.2 → 72.3.
00:00:30	INFO	State Change	State transition: Drift Emerging → Stable. Stable: all integrity domains within acceptable bounds.
00:00:31	WARNING	State Change	State transition: Stable → Drift Emerging. Drift Emerging: Navigation integrity in drift-emerging band (60–75).
00:00:31	WARNING	Threshold Breach	Navigation integrity crossed below threshold 75: 76.8 → 73.0.
00:00:57	WARNING	State Change	State transition: Drift Emerging → Degraded. Degraded: Navigation integrity in degraded band (40–59).
00:00:57	WARNING	Threshold Breach	Navigation integrity crossed below threshold 60: 63.3 → 57.6.

Data Excerpt (Last 15 Steps)

Showing the final 15 time steps of enriched scenario output. Scores are on a 0–100 scale.

Time	State	Nav	Sensor	Comms	GPS Qual	IMU Stab	Wind	Clarity	Latency	Pkt Loss	Batt Stab
00:00:45	Drift Emerging	67.3	84.0	91.0	51.8	85.8	21.7	85.9	13.6	5.3	93.1
00:00:46	Drift Emerging	67.0	88.6	90.0	47.2	91.8	20.7	91.6	11.6	8.7	91.4
00:00:47	Drift Emerging	66.1	83.4	87.8	47.2	86.4	12.0	80.7	16.1	9.0	90.0
00:00:48	Drift Emerging	66.3	85.3	91.5	46.8	87.5	16.9	85.6	15.9	2.5	87.5

Time	State	Nav	Sensor	Comms	GPS Qual	IMU Stab	Wind	Clarity	Latency	Pkt Loss	Batt Stab
00:00:49	Drift Emergi	60.0	90.7	88.5	42.3	93.4	6.1	88.7	15.7	8.1	90.0
00:00:50	Drift Emergi	64.7	87.0	89.4	41.5	88.6	13.3	86.8	17.4	5.1	90.9
00:00:51	Drift Emergi	60.8	83.1	90.9	44.1	86.5	21.0	83.9	14.5	4.7	89.8
00:00:52	Drift Emergi	60.7	87.4	87.0	39.4	82.1	8.3	86.9	18.6	8.5	89.9
00:00:53	Drift Emergi	60.2	81.1	92.7	41.4	88.1	29.5	83.8	10.9	4.3	90.4
00:00:54	Drift Emergi	60.1	82.6	91.6	39.2	87.7	24.0	84.0	13.2	4.5	87.7
00:00:55	Drift Emergi	60.9	84.7	88.0	35.0	86.1	12.0	83.0	16.5	8.3	91.6
00:00:56	Drift Emergi	60.3	86.2	89.7	40.6	90.8	21.3	88.2	10.9	9.9	92.0
00:00:57	Degraded	57.6	84.0	90.2	32.9	84.1	20.5	85.9	14.2	6.2	89.1
00:00:58	Degraded	59.9	88.7	86.6	36.6	83.3	17.0	92.5	20.1	8.0	88.6
00:00:59	Degraded	59.1	85.2	92.6	35.2	85.6	20.7	87.5	14.4	1.7	94.2

Limitations and Non-Goals

Mission Integrity System (MIS) V1 uses simulated telemetry-like data for demonstration and portfolio purposes only. It is not a UAV control system, flight application, autopilot, navigation engine, targeting system, safety-certified tool, predictive model, or production deployment. MIS V1 performs deterministic classification only and does not recommend or automate operational decisions.